Yi Ding

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Google Scholar

Research Interests

Areas Computer Systems, Computer Architecture, Artificial Intelligence (AI), Machine Learning (ML)
Topics AI/ML Systems, Energy-Efficient Computing, Human-Centered Computing, Sustainability

Academic Employment

8/2023-present **Purdue University**, West Lafayette, IN, USA
Assistant Professor in Elmore Family School of Electrical and Computer Engineering
Affiliated Faculty, Institute for a Sustainable Future (ISF)
Affiliated Faculty, Institute for Physical Artificial Intelligence (IPAI)

1/2021-8/2023 **Massachusetts Institute of Technology**, Cambridge, MA, USA
Postdoctoral Associate & NSF Computing Innovation Fellow. Mentor: Michael Carbin.

Education

2015–2020 **The University of Chicago**, Chicago, IL, USA
PhD & MS in Computer Science. Advisor: Henry Hoffmann
PhD Dissertation: *Learning Structure for Computer Systems Management*

2013–2015 **Nanyang Technological University**, Singapore
Doctoral Student in Computer Science. Passed Qualification Exam.

2008–2012 **Beijing Jiaotong University**, Beijing, China
B.E. in Electronic Science and Technology. Graduated with *Highest Honor*.

Industry Employment

10/2021-12/2022 **Meta**, Cambridge, MA, USA
Visiting Researcher. Led a team to improve maintenance efficiency of servers in hyperscale datacenters.

6/2019-9/2019 **Google**, Sunnyvale, CA, USA
Research Intern. Introduced causal analysis to fleet understanding and automated confounder discovery.

Awards and Honors

2023 Innovation Award, Quantum Computing for Drug Discovery Challenge at ICCAD
2020-2023 NSF/CRA/CCC Computing Innovation Fellowship [Link].
2021 Meta Research Award [Link]
2020 EECS Rising Stars at UC Berkeley [Link]

Publications

Underline Students advised by me; ⋆ Equal contribution; † Corresponding faculty author

MLSys'26 **Cost-aware Duration Prediction for Software Upgrades in Datacenters** [DOI]
Yi Ding†, Aijia Gao, Thibaud Ryden, Michal Sedlak, Essam Ewaisha, Igor Marnat, Henry Hoffmann.
The 9th Annual Conference on Machine Learning and Systems, 2026.

ICLR'26 **ViTSP: A Vision Language Models Guided Framework for Large-Scale Traveling Salesman Problems** [DOI]
Zhuoli Yin, **Yi Ding**, Reem Khir, Hua Cai.
The 14th International Conference on Learning Representations, 2026.

SIGMETRICS'26 **Cache Your Prompt When It's Green — Carbon-Aware Caching for Large Language Model Serving** [DOI]
Yuyang Tian, Desen Sun, **Yi Ding**, Sihang Liu.
ACM on Measurement and Analysis of Computer Systems, 2026.

CHASE'26 **From Detection to Forecasting: An Empirical Study of College Student Mental Health Using Smartphone Sensing**

Jheel Kishore Gala*, Asher Sprigler*, Priyanshu Rajeshbhai Malaviya, Elizabeth Shen, Yixue Zhao, **Yi Ding**†, Xipeng Shen†.

IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies, 2026.

CHASE'26 **Demo: Trivana: On-Device Multimodal Forecasting and LLM-Driven Mental Health Support**

Asher Sprigler, Priyanshu Rajeshbhai Malaviya, Elizabeth Shen, Jheel Kishore Gala, Tanay Taral Shah, Yixue Zhao, **Yi Ding**†, Xipeng Shen†.

IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies, 2026.

UMAP'26 LBR **Personalization Matters: Benchmarking ML, DL and LLMs for Mental Health Forecasting from Smartphone Sensing Data**

Kaidong Feng, Zhu Sun, Roy Ka-Wei Lee, Xun Jiang, Yin-Leng Theng, **Yi Ding**.

The 34th ACM International Conference on User Modeling, Adaptation and Personalization, 2026.

IEEE CAL'25 **Disaggregated Speculative Decoding for Carbon-Efficient LLM Serving** [DOI]

Tianyao Shi*, Yanran Wu*, Sihang Liu, **Yi Ding**†.

IEEE Computer Architecture Letters, 2025.

ACL'25 **Unveiling Environmental Impacts of Large Language Model Serving: A Functional Unit View** [DOI]

Yanran Wu, Inez Hua, **Yi Ding**†.

The 63rd Annual Meeting of the Association for Computational Linguistics, 2025.

HotCarbon'25 **Not All Water Consumption Is Equal: A Water Stress Weighted Metric for Sustainable Computing** [DOI]

Yanran Wu, Inez Hua, **Yi Ding**†.

ACM SIGENERGY Energy Informatics Review (EIR) and Workshop on Sustainable Computer Systems, 2025.

HotCarbon'25 **When Servers Meet Species: A Fab-to-Grave Lens on Computing's Biodiversity Impact** [DOI]

Tianyao Shi, Ritbik Kumar, Inez Hua, **Yi Ding**†.

ACM SIGENERGY Energy Informatics Review (EIR) and Workshop on Sustainable Computer Systems, 2025.

e-Energy'25 **EnsembleCI: Ensemble Learning for Carbon Intensity Forecasting** [DOI]

Leyi Yan, Linda Wang, Sihang Liu, **Yi Ding**†.

The 16th ACM International Conference on Future Energy Systems, 2025.

CHASE'25 **Predicting and Understanding College Student Mental Health with Interpretable Machine Learning** [DOI]

Meghna Roy Chowdhury, Wei Xuan, Sheyras Sen, Yixue Zhao, **Yi Ding**†.

IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies, 2025.

MOBILESoft'25 **Unlocking Mental Health: Exploring College Students' Well-being through Smartphone Behaviors** [DOI]

Wei Xuan, Meghna Roy Chowdhury, **Yi Ding**, Yixue Zhao.

IEEE/ACM 12th International Conference on Mobile Software Engineering and Systems, 2025.

EMBC'25 **SSL-SE-EEG: A Framework for Robust Learning from Unlabeled EEG Data with Self-Supervised Learning and Squeeze-Excitation Networks**

Meghna Roy Chowdhury, **Yi Ding**, Shreyas Sen.

The 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society, 2025.

EM'25 **Beyond Climate Change: A Holistic Framework for Evaluating the Environmental Impact of Computing Systems**

Yi Ding†, Tianyao Shi, Yanran Wu, Inez Hua.

The Magazine for Environmental Manager, 2025.

Dagstuhl'25 **Climate Change: What is Computing's Responsibility?** [DOI]

Bran Knowles, Vicki L. Hanson, Christoph Becker, Mike Berners-Lee, Andrew A. Chien, Benoit Combemale, Vlad Coroama, Koen De Bosschere, **Yi Ding**, Adrian Friday, Boris Gamazaychikov, Lynda Hardman, Simon Hinterholzer, Mattias Hojer, Lynn Kaack, Lenneke Kuijer, Anne-Laure Ligozat, Jan Tobias Muehlberg, Yunmook Nah, Thomas Olsson, Anne-Cecile Orgerie, Daniel Pargman, Birgit Penzenstadler, Tom Romanoff, Emma Strubell, Colin Venters, Junhua Zhao.

Dagstuhl Perspectives Workshop 25122. In Dagstuhl Reports, Volume 15, Issue 3, pp. 113-124, 2025.

IGSC'24 **Sustainable LLM Serving: Environmental Implications, Challenges, and Opportunities** [DOI]
Yi Ding†, Tianyao Shi.
The 15th International Green and Sustainable Computing Conference, 2024.

HotCarbon'24 **Uncertainty-Aware Decarbonization for Datacenters** [DOI]
Amy Li, Sihang Liu, **Yi Ding**†.
ACM SIGENERGY Energy Informatics Review (EIR) and Workshop on Sustainable Computer Systems, 2024.

HotCarbon'24 **Towards Sustainable Large Language Model Serving** [DOI]
Sophia Nguyen*, Beihao Zhou*, **Yi Ding**, Sihang Liu.
ACM SIGENERGY Energy Informatics Review (EIR) and Workshop on Sustainable Computer Systems, 2024.

ASPLOS'23 **CAFQA: A Classical Simulation Bootstrap for Variational Quantum Algorithms** [DOI]
Gokul Subramanian Ravi, Pranav Gokhale, **Yi Ding**, William Kirby, Kaitlin Smith, Jonathan M. Baker, Peter J. Love, Henry Hoffmann, Kenneth R. Brown, Frederic T. Chong.
The 28th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, 2023.
2023 Innovation Award, Quantum Computing for Drug Discovery Challenge at ICCAD.

OOPSLA'23 **Turaco: Complexity-Guided Data Sampling for Training Neural Surrogates of Programs** [DOI]
Alex Renda, **Yi Ding**, Michael Carbin.
ACM SIGPLAN Conference on Object-Oriented Programming, Systems, Languages, and Applications, 2023.

Biometrika'23 **Minimax Designs for Causal Effects in Temporal Experiments with Treatment Habituation** [DOI]
Guillaume W Basse, **Yi Ding**, Panos Toulis.
Biometrika, 2023.

MLSys'22 **NURD: Negative-Unlabeled Learning for Online Datacenter Straggler Prediction**
Yi Ding, Avinash Rao, Hyebin Song, Rebecca Willett, Henry Hoffmann.
Machine Learning and Systems, 2022.

ESEC/FSE'21 **Generalizable and Interpretable Learning for Configuration Extrapolation** [DOI]
Yi Ding, Ahsan Pervaiz, Michael Carbin, Henry Hoffmann.
The 29th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering, 2021.

Onward!'21 **Programming with Neural Surrogates of Programs** [DOI]
Alex Renda, **Yi Ding**, Michael Carbin.
ACM SIGPLAN International Symposium on New Ideas, New Paradigms, and Reflections on Programming and Software, 2021.

npj Urban'21 **Neighborhood Street Activity and Greenspace Usage Uniquely Contribute to Predicting Crime** [DOI]
Kathryn E. Schertz, James Saxon, Carlos Cardenas-Iniguez, Luis Bettencourt, **Yi Ding**, Henry Hoffmann, Marc G. Berman.
npj Urban Sustainability, 2021.

AISTATS'20 **Dynamical Systems Theory for Causal Inference with Application to Synthetic Control Methods** [DOI]
Yi Ding, Panos Toulis.
International Conference on Artificial Intelligence and Statistics, 2020.

NeurIPS'20 **A Polynomial-Time Algorithm for Learning Nonparametric Causal Graphs** [DOI]
Ming Gao, **Yi Ding**, Bryon Aragam.
The 34th Annual Conference on Advances in Neural Information Processing Systems, 2020.

ISCA'19 **Generative and Multi-Phase Learning for Computer Systems Optimization** [DOI]
Yi Ding, Nikita Mishra, Henry Hoffmann.
The 46th International Symposium on Computer Architecture, 2019.

NeurIPS'17 **Multiresolution Kernel Approximation for Gaussian Process Regression** [DOI]
Yi Ding, Risi Kondor, Jonathan Eskreis-Winkler.
The 31st Annual Conference on Advances in Neural Information Processing Systems, 2017.

Spotlight Presentation, Top 4% of Submissions.

ICDM'17 **Large Scale Kernel Methods for Online AUC Maximization** [DOI]
Yi Ding, Chenghao Liu, Peilin Zhao, Steven C.H. Hoi.
IEEE International Conference on Data Mining, 2017.
Long Oral, Top 8% of Submissions.

AAAI'15 **An Adaptive Gradient Method for Online AUC Maximization** [DOI]
Yi Ding, Peilin Zhao, Steven C.H. Hoi, Yew-Soon Ong.
The 29th AAAI Conference on Artificial Intelligence, 2015.
Oral Presentation, Top 10% of Submissions.

AAAI'14 **Learning Relative Similarity by Stochastic Dual Coordinate Ascent** [DOI]
Pengcheng Wu, **Yi Ding**, Peilin Zhao, Chunyan Miao, Steven C.H. Hoi.
The 28th AAAI Conference on Artificial Intelligence, 2014.

Grants

2024-2026
Funder: National Science Foundation
Title: Conference: DESC: Type III: A Holistic AI Computing Framework: Incorporating the Water and Biodiversity Dimensions of Sustainability
People: Inez Hua (PI), Yi Ding
Awarded: $99,992 (my share: 50%)

2024-2025
Funder: Purdue University
Title: Seed Funding for High-Impact Review Papers
People: Inez Hua (PI), Yi Ding
Awarded: $10,000 (my share: 50%)

2021-2022
Funder: Meta
Title: Negative-Unlabeled Learning for Online Datacenter Straggler Prediction
People: Michael Carbin (PI), Yi Ding
Awarded: $46,000

2020-2023
Funder: National Science Foundation
Title: CRA/CCC Computing Innovation Fellowship
People: Yi Ding
Awarded: $295,704

Teaching

Instructor, Purdue University

Fall 2026 Systems for AI and Large Language Models (ECE 69500)
Spring 2026 Python for Data Science (ECE 20875)
Fall 2025 Machine Learning in Cloud Computing (ECE 69500)
Spring 2025 Python for Data Science (ECE 20875)
Fall 2024 Machine Learning in Cloud Computing (ECE 69500)
Spring 2024 Python for Data Science (ECE 20875)
Fall 2023 Python for Data Science (ECE 20875)

Research Advising

PhD Students

Fall 2025- Asher Sprigler (ECE), Purdue University
Fall 2025- Lauren Caccamise (ECE), Purdue University
Fall 2024- Tianyao Shi (ECE), Purdue University

PhD Thesis Committee

Fall 2021- Meghna Roy Chowdhury (ECE), Purdue University
Fall 2021- Zhuoli Yin (IE), Purdue University

MS Thesis Committee

Fall 2025- Taewoong Yoon (ECE), Purdue University
Fall 2025- Daeun Oh (ECE, Chair), Purdue University

Fall 2023-2025 Amir Massah Bavani (ECE), Purdue University

MS Students

2021-2022 Hyunji Kim, MIT

Undergraduate Students

Summer 2025 Barbara Su, Rice University (Remote Intern, Purdue SURF)

Spring 2025 Jaewon Cho, Purdue University (DUIRI, awarded $1,000 fellowship)

Spring 2025 Isha Shamim, Purdue University (DUIRI, awarded $1,000 fellowship)

Fall 2024 Gavin Fortwendel, Purdue University (DUIRI, awarded $1,000 fellowship; Won 1st Place in Research Talk, College of Engineering, Fall 2024 Undergraduate Research Expo)

Fall 2024 Sarah Deniz, Purdue University (DUIRI, awarded $1,000 fellowship)

Fall 2024 Leyi Yan, University of Waterloo (Co-author, e-Energy 2025 paper)

Fall 2024 Linda Wang, University of Waterloo (Co-author, e-Energy 2025 paper)

Spring 2024 Amy Li, University of Waterloo (Co-author, HotCarbon 2024 paper)

Spring 2024 Beihao Zhou, University of Waterloo (Co-author, HotCarbon 2024 paper)

Spring 2024 Sophia Nguyen, University of Waterloo (Co-author, HotCarbon 2024 paper)

2019-2020 Avinash Rao, University of Chicago (Co-author, MLSys 2022 paper)

Professional Service

Organizer

2026 Program Committee Co-Chair, ACM Workshop on Sustainable Computer Systems (HotCarbon) 2026

2026 Co-Chair, HARMONY Workshop on AI and Mental Health, co-located with IEEE/ACM CHASE 2026

2026 Co-Chair, The 1st Workshop on Trustworthy and Adaptive LLMs for Mental and Physical Wellbeing in Recommendations, co-located with ACM UMAP 2026

2024 Co-Chair, NSF Workshop on Sustainable Computing: AI, Water, and Biodiversity

Invited Participant

2026 NII Shonan Meeting: Green Intelligence: Sustainable Development and Operations of Intelligent Systems

2026 CRA Career Mentoring Workshop

2026 DOE/AFOSR Energy Consequences of Information Workshop

2025 Indiana Water Summit

2025 CCC Computing Futures Symposium

2025 CIFellows 2025 Symposium

2025 Dagstuhl Perspectives Workshop: Climate Change: What is Computing's Responsibility?

2024 NSF Workshop on Sustainable Computing for Sustainability

2022 NITRD 30th Anniversary Symposium

2022 CIFellows 2022 Workshop

Program Committee

2027 IEEE International Symposium on High-Performance Computer Architecture (HPCA)

2026 IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS)

2026 ACM International Conference on Future and Sustainable Energy Systems (e-Energy)

2026 ACM Intl. Conf. on Architectural Support for Programming Languages and Operating Systems (ASPLOS)

2025 IEEE Computer Architecture Letters (CAL)

2025 ACM Workshop on Sustainable Computer Systems (HotCarbon)

2025 SIGOPS Asia-Pacific Workshop on Systems (APSys)

2025 IEEE/ACM International Symposium on Computer Architecture (ISCA)

2025 IEEE International Symposium on High-Performance Computer Architecture (HPCA)

2024 USENIX Annual Technical Conference (ATC)

2024 Conference on Systems and Machine Learning (MLSys)

2023 ACM Student Research Competition at PACT

2022 ACM SPLASH Onward!

2022 Conference on Systems and Machine Learning (MLSys)

2022 ACM Asia-Pacific Workshop on Systems

2022 Journal of Systems Research

Grant Review

2026 NSF Panel, NSF Ad Hoc

2025 Huo Family Foundation

2024 NSF GRFP

Technical Reviewing

2026 Transactions on Machine Learning Research (TMLR)

2026 International Conference on Learning Representations (ICLR)

2022 Neural Information Processing Systems (NeurIPS)

2022 International Conference on Learning Representations (ICLR)

2022 International Conference on Machine Learning (ICML)

2021 Neural Information Processing Systems (NeurIPS)

2021 AAAI Conference on Artificial Intelligence (AAAI)

2020 AAAI Conference on Artificial Intelligence (AAAI)

2019 Neural Information Processing Systems (NeurIPS)

2019 International Conference on Machine Learning (ICML)

Presentations

Invited Talks

2026 Rethinking AI: From Energy Efficiency to Environmental Intelligence
- 9Zero Climate Innovation Hub, San Francisco, USA — Apr. 2026

2026 Demystifying the Carbon Emissions of AI Datacenters
- Hoosier Environmental Council, Webinar, Virtual — Mar. 2026

2026 Systematic Characterization of LLM Quantization: A Performance, Energy, and Quality Perspective
- DOE/AFOSR Energy Consequences of Information Workshop, Santa Fe, USA — Feb. 2026

2025 Not All Water Consumption Is Equal: A Water Stress Weighted Metric for Sustainable Computing
- Green Software Foundation (Virtual) — Sep. 2025

2025 Not All Water Consumption Is Equal: An AI, Datacenter, and Semiconductor Perspective
- Indiana Water Summit, Indianapolis, USA — Aug. 2025

2024 Towards Sustainable Next Generation AI and Cloud Systems
- Meta, Sunnyvale, USA — Sep. 2024

2024 Uncertainty-Aware Carbon Optimization in Cloud Computing
- SoDec Workshop at e-Energy, Singapore — Jun. 2024
- ASTAR Center for Frontier AI Research (CFAR), Singapore — Jun. 2024

2023 A Holistic View on Machine Learning for Systems
- University of Waterloo, Department of Computer Science — Jun. 2023
- Microsoft Research — Apr. 2023
- Texas A&M University, Department of Computer Science and Engineering — Apr. 2023
- University of Southern California, Department of Electrical and Computer Engineering — Apr. 2023
- University of Illinois, Department of Computer Science — Mar. 2023
- Cornell Tech, Department of Electrical and Computer Engineering — Mar. 2023
- Washington University in St. Louis, Department of Computer Science & Engineering — Mar. 2023
- Purdue University, Department of Electrical and Computer Engineering — Mar. 2023
- Purdue University, Department of Computer Science — Mar. 2023
- Virginia Tech, Department of Computer Science — Mar. 2023
- Indiana University Bloomington, Department of Computer Science — Feb. 2023
- University of Colorado Boulder, Department of Computer Science — Feb. 2023
- University of Massachusetts Amherst, College of Information and Computer Sciences — Feb. 2023

Conference Presentations

2026 Cost-aware Duration Prediction for Software Upgrades in Datacenters
- Conference presentation at MLSys, Bellevue, USA — May. 2026

2024 Sustainable LLM Serving: Environmental Implications, Challenges, and Opportunities
- Conference presentation at IGSC, Austin, USA — Oct. 2024

2024 Uncertainty-Aware Decarbonization for Datacenters
- Conference presentation at HotCarbon, Santa Cruz, USA — Jul. 2024

2022 NURD: Negative-Unlabeled Learning for Online Datacenter Straggler Prediction
- Conference presentation at MLSys, Santa Clara, USA — Aug. 2022

2022 Predictable Maintenance Job Planning in Datacenters
- Meta Infrastructure Data Science Faculty Workshop at KDD, DC, USA — Aug. 2022

2021 Generalizable and Interpretable Learning for Configuration Extrapolation
- Conference presentation at ESEC/FSE, Virtual — Nov. 2021

2020 Dynamical Systems Theory for Causal Inference with Application to Synthetic Controls
- Causal Data Science Meeting, Virtual — Nov. 2020
- Conference presentation at AISTATS, Virtual — Aug. 2020

2019 Generative and Multi–phase Learning for Computer Systems Optimization
- Conference presentation at ISCA, Phoenix, USA — Jun. 2019

2017 Multiresolution Kernel Approximation for Gaussian Process Regression
- Conference presentation at NeurIPS, Long Beach, USA — Dec. 2017

2017 Large Scale Kernel Methods for Online AUC Maximization
- Conference presentation at ICDM, New Orleans, USA — Nov. 2017

2015 An Adaptive Gradient Method for Online AUC Maximization
- Conference presentation at AAAI, Austin, USA — Jan. 2015

Media Coverage

3/31/2026 Texas Water Strain Raises Questions for Data Center Boom. Bloomberg Government

1/21/2026 The Hidden Cost of AI: Making Data Centers Sustainable. Purdue ECE Podcast

11/24/2025 How to make data centers less thirsty. Grist

10/15/2025 Purdue ECE research reveals how computing impacts global biodiversity. Purdue ECE News

10/02/2025 Economic boom or environmental disaster? Rural Texas grapples with pros, cons of data centers. The Texas Tribune

09/25/2025 Data centers are thirsty for Texas' water, but state planners don't know how much they will need. The Texas Tribune

09/16/2025 Spain's data centre law: supporting local groups in the public consultation. The Green Web Foundation

08/15/2025 Big Tech's big thirst — AI's demand for Texas water. Texas Public Radio

07/15/2025 Companies focus on ways of achieving energy efficiency as consumption keeps increasing. New Zealand Herald

07/14/2025 Tech Giants Scramble To Meet AI's Looming Energy Crisis. Barron's

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